FIRST LESSON

The Most Beautiful of Theories

In his youth Albert Einstein spent a year loafing aimlessly. You don't get anywhere by not 'wasting' time – something, unfortunately, which the parents of teenagers tend frequently to forget. He was in Pavia. He had joined his family having abandoned his studies in Germany, unable to endure the rigours of his high school there. It was the beginning of the twentieth century, and in Italy the beginning of its industrial revolution. His father, an engineer, was installing the first electrical power plants in the Paduan plains. Albert was reading Kant and attending occasional lectures at the University of Pavia: for pleasure, without being registered there or having to think about exams. It is thus that serious scientists are made.

After this he registered at the Polytechnic Institute in Zurich, and immersed himself in the study of physics. A few years later, in 1905, he sent three articles to the most prestigious scientific journal of the period, the *Annalen der Physik*. Each of these is worthy of a Nobel Prize. The first shows that atoms really exist. The second lays the first foundation for quantum mechanics, which I will discuss in the next lesson. The third presents his first theory of relativity (known today as 'special relativity'), the theory which elucidates how time does not pass identically for everyone: two identical twins find that they are different in age, if one of them has travelled at speed.

Einstein became a renowned scientist overnight and received offers of employment from various universities. But something disturbed him: despite its immediate acclaim, his theory of relativity did not fit with what we know about gravity, namely with how things fall. He came to realize this when writing an article summarizing his theory, and began to wonder if the law of 'universal gravity' as formulated by the father of physics himself, Isaac Newton, was in need of revision in order to make it compatible with the new concept of relativity. He immersed himself in the problem. It would take ten years to resolve. Ten years of frenzied study, attempts, errors, confusion, mistaken articles, brilliant ideas, misconceived ideas.